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Image Tampering Detection Using Error Level Analysis and Metadata Analysis

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***Abstract*– As the volume of digital content continues to increase, the problem of information pollution has become increasingly pressing. Information pollution refers to the dissemination of false, misleading, or biased information through various channels such as social media, news outlets, and websites. This can include fake news, propaganda, as well as tampered or edited images and videos. With the proliferation of digital image editing software and artificial intelligence (AI) techniques, it is increasingly challenging to determine the originality of a photograph. Field of digital image forensics, which aims to identify and prevent image forgery, has gained significant attention in last few years. In this study, we propose a combined approach of Error Level Analysis (ELA) and Metadata Analysis for the passive authentication of images. Utilizing deep neural networks, our proposed approach seeks to evaluate the originality of an image. System demonstrated a test accuracy of 87.24% for the ELA approach and a test accuracy of 73% for the metadata analysis approach.**

Keywords*– *Image forgery, Error Level Analysis, Metadata Analysis, CNN, Deep Learning

I. INTRODUCTION

Detecting alterations in digital images has become a significant area of study in recent years, as the authenticity of digital photos is increasingly important in a world where more than 3.2 billion images are uploaded to the internet daily. Image forgery is a very significant problem because it can be used to spread misinformation, sway public opinion, and even perpetrate crimes. Image forgery detection algorithms are a useful tool for identifying and preventing the spread of tampered images, and for protection against the negative impacts of image forgery. It is also more cost-effective than hiring additional staff or contracting with external experts to manually inspect images.

II. LITERATURE REVIEW

Keypoints are used to detect image alterations [1]. Features extracted from rectangular splits of the image are used to infer any connected regions, deciding if the image is altered. Study [2] proposes a CNN-based approach to extract and employ unique camera footprints in iterative clustering to analyze the image. Feature-based approach with camera parameters to detect the forgery is combined in [3]. Features extracted from the image whose camera is known are fed into the multivariate gaussian model. Log-likelihood is used to evaluate forgeries. Study [4] comprises an in-depth analysis of the deep learning-based methods to detect passive forgery. Datasets currently being used for the application are discussed. In order to identify fake images, two techniques have been combined [5]. Block matching and residual-based local descriptors are employed to identify splicing and copy-move forgery.

To identify splicing and copy-move forgeries, a deep learning-based method is suggested in [6]. The model is trained using RGB images to learn the hierarchical representations. CNN-based Camera model identification (CMI) is done to be used with a global tampering detection framework [7]. The image is considered altered if multiple CMI is found for a single image. Block-based and feature-based forgery detection are fused together in [8]. FAST detectors are used for keypoint extraction and Brute-Force feature matching is done for matching the blocks. Motion Blur based detection is done using CNNs [9]. Motion blur kernels are employed for tamper detection. Encoder-Decoder architecture has been proposed to detect and localize image tampering [10]. Dense Fully CN is trained using a custom dataset, made using photoshop scripting which imitates forgery.

Local and Global Tampering Detection is done using Color Filter Array (CFA) demosaicing [11]. Researchers have used the fact that upon image alteration, CFA demosaicing changes significantly. In [12] CNN-based frameworks that detect face manipulations are proposed. The framework is trained using a balanced dataset, while another, hybrid method while trained with an imbalanced dataset, is boosted with Gradient and AdaBoost. In [13] authors have proposed a method for forgery detection using camera model identification (CMI). Image transformation and patch extraction are done to detect the camera footprints using CNN and the results are analyzed to classify whether an image is altered or not.

A deep learning-based semantic image segmentation network is used to detect forgery [14]. Detection of Splicing in Modified Blurred images [15] correctly categorises image regions with motion blur and out-of-focus effects. Partial recovery of the warped sections is carried out in [17], and the position of the distorted regions is determined by changes made to the image's content. Evaluation of tampered images in image forensics is detected using error level analysis (ELA) [18]. Study shows that ELA can be applicable to any image if no resizing operation has been performed. A comprehensive study of various image forgery methods is discussed in [19]. The detection processes used by the methods with the best accuracy are extremely complicated.

Metadata analysis and integrity of the double compression are used to Detect video forgery [20]. This approach is utilized to identify tampering, including frame insertion, manipulation of the fading in and out, and frame deletion only. In [21], a method for copy-paste forgery detection using

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a Histogram of Oriented Gradients (HOG) is proposed. Comparative studies on various copy-move forgery techniques are discussed in [22]. Feature matching techniques have the limitation of determining accurate values. A suggested forgery detection approach uses threshold and contouring techniques [23] to identify the fabricated part and is based on similarities between the images. The structural similarity index (SSIM) and CNN are used in the technique to identify fake photos.

In [24], a reliable and effective method of detecting fake images is suggested. The work uses SIFT feature extraction algorithm and Stationary Wavelet Transform (SWT) decomposition strategy to detect forgeries. Over a typical dataset of images, it achieved accuracy of 93%. A five-stage based copy-move forgery approach [25] was proposed and the forged region was detected successfully with an accuracy of 98.44 %. The study focuses on JPEG format and identifies two forgeries using copy-move and splicing forgeries [26]. The first approach employs DCT coefficients using double JPEG compression, while another uses feature key-points via SURF descriptors.

Metadata is a prominent factor in image forensics. One of the earliest studies [27] used metadata for image retrieval. Metadata is combined with CNN [28] for predicting geological factors such as temperature, humidity, sun-altitude angle, and weather. Geological information is used for authenticity from publicly available sources. Different trust models combined with metadata analysis are used to obtain various assessments [29]. ELA with JPEG Compression is used to identify modifications while Noise analysis is used to identify the affected area. Video tampering is detected using the metadata with machine learning [30], authors have focused on the .mp4 container to use its tree structure. Splicing in satellite images is detected using GANs [31]. Popular pix2pix architecture is used. Deep Neural networks have been trained to detect image tampering [32]. ELA pre-processing is done and for the training, CASIA 2.0 Dataset is used. A comprehensive study of image tampering is done in [33]. Current trends to detect image forgery and aspect of GANs with image forensics has been thoroughly discussed. Double compression images have a DCT matrix, whose histogram shows a peak if any tampering is done [34]. Study has proposed an algorithm to detect tampering on Double Compressed images.

In [35], a comprehensive examination was conducted on the detection of passive digital image forgery. The study provided a thorough analysis of copy-move forgery and the corresponding detection algorithms. Self-supervised-based algorithms have been proposed by the authors to detect image forgery [36]. EXIF metadata has been used by authors as a training supervisory signal. A classification neural network has been trained to differentiate between the image and the mirror of the image, aiming to solve the problem of chirality [37]. The study adopted RESNET architecture treating this as a problem of standard classification.

The study provides a comprehensive review of universal image forgery [38]. Different detection methods have been discussed in depth. In [39], the proposed method focuses on classifying whether a certain image posted on a popular social platform is fake or not. The algorithm traces images and their caption's online presence and uses the inferred data on BI-directional LSTM to obtain the end result.

III. METHODOLOGY

Study employs two parallel verticals to detect the forgery on an image - Error Level Analysis (ELA) and Metadata Analysis. ELA approach exploits CNN to detect the forgery. Dataset of Real and Fake images used for preprocessing. ELA is performed on each sample of the Real and Tampered images, producing a new dataset which contains error level analysis, categorized for Real and Tampered Images. We train our model using a newly obtained dataset of ELAs, to learn to classify between an untampered and forged image. An image which is to be checked for forgery is first passed through Error Level Analysis and then is checked by the model for any potential tampering.

The proposed method exploits metadata analysis to give more insight into the authenticity of an image. Extracted metadata is used to get the weather at a given date and time of a location image is tagged with. Trained Weather Classifier CNN predicts the weather in an image, which now can be cross-checked with the original weather condition obtained using metadata. The system overview is depicted in Fig. 1.

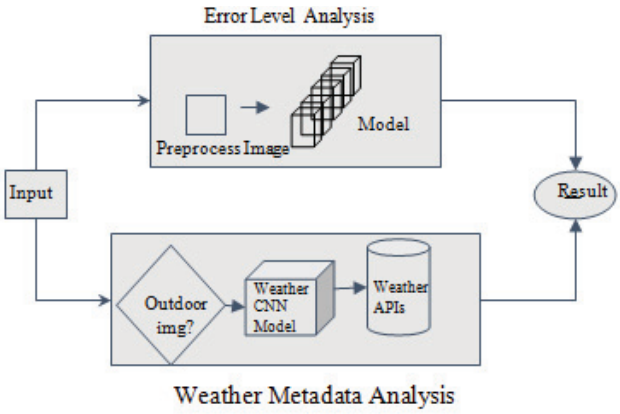

Fig. 1. Block Diagram of the System.

A. ELA - Error Level Analysis

Error Level Analysis helps in identifying the areas which have been altered in a lossy compression image. The difference is noticeable at different compression levels. By using JPEG compression, an image should have the same level throughout if it hasn't been altered. If a certain area shows a high level of error analysis, then it most likely indicates that it is gone through digital manipulations.

ELA emphasises the variations in JPEG compression levels. There are several factors which can help to check whether an image has been tampered with or not:

1. Regions with uniform colouring (eg. Solid wall or sky), should have lower ELA (darker).
2. Similar edges usually have similar brightness. When it comes to appearance, high-contrast edges will resemble other high-contrast edges, whereas low-contrast edges resemble other low-contrast edges.
3. Similar textures usually exhibit similar colouring.
4. Regardless of the original colour, all flat surfaces have the same colouring.

A lossy compression like JPEG uses a specific quality level before saving the picture. As we lower the quality level, JPEG compresses an image by removing some colour information and we get a higher compression ratio. This

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might visually look similar to the human eye, but at the pixel level, values are different. The JPEG algorithm uses an 8x8 pixel grid. All the 8x8 squares are compressed separately. The error level for each 8x8 square should be similar if the image hasn't been altered in any way. Every square should degrade roughly at the same rate if the image is left unaltered and resaved with lower quality.

Error Level Analysis for a real image has been done multiple times, lowering 5 units of quality level with each re-saves. ELA at initial re-saves shows a red and blue tint, this is because JPEG uses luminance and chromatic channels (red and blue) to save an image. In this case of a real image, the edges have similar brightness, the uniform region of the river has lower ELA and the flat surface as well appears to have similar colouring.

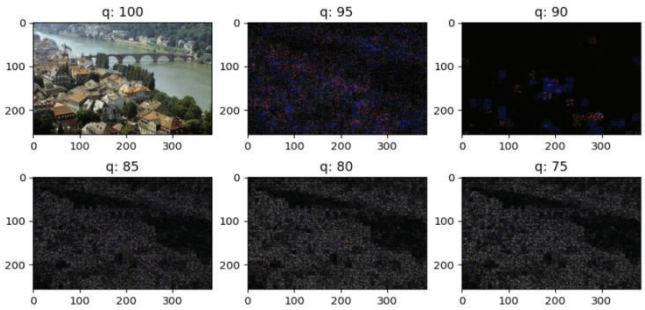

Fig. 2. (a). ELA - Real Image

Error Level Analysis for an altered image has been done below, again lowering 5 units of quality level with each resaves. As we lower the quality level, there isn't much of a change in ELA, but at initial levels, we can see that the high-contrast edges are quite distinctive from each other in terms of brightness. This indicates that the image might have gone under digital modification.

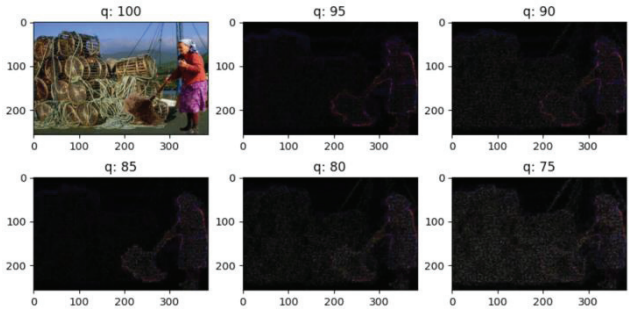

Fig. 2. (b). ELA - Tampered Image

A comparative example of ELA of real and the same image with modifications at a quality of 95 is shown below:

Fig. 3. (a). Real Image

ELA shows low error as the majority of the surface is uniform background. Some edges are highlighted but have similar brightness compared with edges of similar contrast.

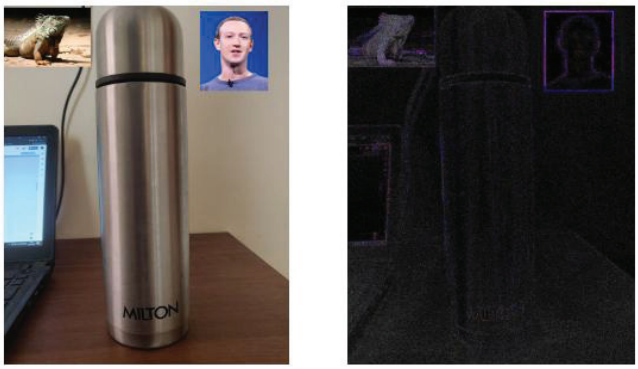

Fig. 3.(b). Tampered Image

ELA highlights the tampered area, with edges remarkably having distinctive brightness at the same level.

B. *Metadata Analysis*

EXIF, Exchangeable Image file, is a standard format used by almost every digital camera to store interchange information (metadata). EXIF format uses .jpeg compression to store the annotations. Every digital camera uses EXIF annotations to store the metadata. Metadata details the important information about the image in form of attributes. These attributes can be dimension, captured date, time, location, bit depth, focal length, shutter speed etc. Some of the common attributes a digital image can have are as follow

TABLE I.

Date	Location
Dimensions	Camera Model
ISO Speed	Compression
Exposure	Aperture
Bit Depth	White Balance

C. *Datasets*

1) *ELA Preprocessing*

We have created a new dataset for our model using CASIA V2 Dataset for image forgery. The dataset contains around 7492 authentic images and 5125 tampered images. Some of these are redundant as ELA requires lossy compression(.jpeg/.jpg), so to make the data balanced, we have dropped some samples, preserving them for testing later. The final dataset we had for ELA preprocessing and training the model was of 2437 Real images and 2056 n Tampered images. Additionally, it also has the ground truth of all the tampered images. Class proportions are shown in the following table:

TABLE II.

Class	**No of Images**	**Percentage**
Authentic	2437	54.2%
Tampered	2056	45.8%

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2) Metadata Analysis based Weather CNN

We have obtained the dataset from various sources such as Image2Weather Dataset, Kaggle, etc. We have selected only some numbers of images from the selected class. The images we have filtered are the ones with less noise which depicts the weather condition clearly and helps the training model to be more generalised towards testing images. Class proportions are shown in the table below:

TABLE III.

Class	No of Images	Percentage
Lightning	377	16.7%
Rainy	526	23.33%
Cloudy	620	27.5%
Sunny	732	32.5%

D. Design and Implementation Of The System

1) ELA

The Working for ELA approach has been depicted in the following block diagram.

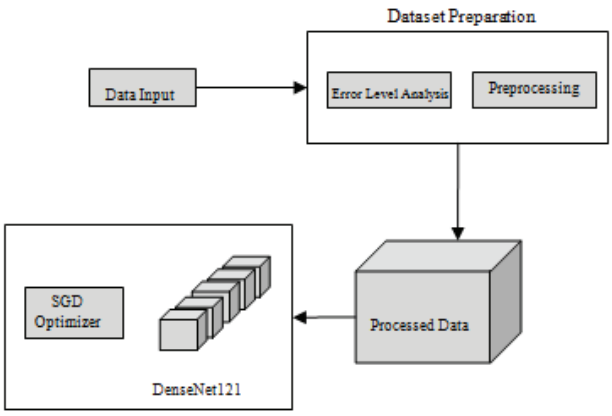

Fig. 4. Error Level Analysis Approach

E. ELA Preprocessing

All the images of class authentic and tampered is preprocessed to produce a new dataset containing ELA for each image. ELA refers to the difference between the varying compression level of images at different quality levels. We use PIL Module and image channel operations to perform error-level analysis on an image.

Firstly, an original image is re-saved with JPEG encoding at a certain constant quality level (Q). Absolute difference is determined between the original and this resaved image, and this new output of difference between the two compression levels presents error level analysis.

The initial results we obtain are much darker for uniform backgrounds, hence it is ideal to enhance the brightness before feeding this data to train. The scale by which the image should be brightened (S) is calculated optimally. Procedural Steps for Error Level Analysis of an image are shown in Algorithm 1.

Algorithm 1 ELA

Input: Image
Output: Error Level Analysis of Image

```
temp = image.copy()
save(temp,comression_quality = 90, name = saved_img)
ela_image = abs_difference(saved_img,image)
Calculate the extrema for R, G and B channels.
        extremas = get_extrema(ela_image)
max_diff = sum([for ex in extrema])/3
if max_dif==0:
        max_dif =1
scale = 255.0/max_diff
ela_image.brighten(scale)
return ela_image
```

The value of compression quality level Q is subjective. Empirically, we found that Q between 90-95 yields the most optimal results, retaining good contrast and highlighting tampered areas if any. We have used Q=95 for ELA preprocessing.

Iteratively, the algorithm is repeated till we get a new dataset, containing ELA of the Real and Tampered images.

F. Model Architecture

Utilizing images as training data, the feed-forward network CNN can infer various categorization tasks. While real and tampered images can be fed into the CNN model for training as is, the intuition behind using ELA is that it makes it easier to analyze and learn to identify any manipulations. The dataset of Real and Tampered images is first processed to produce a new dataset which has ELA of Authentic and Tampered images. New dataset is trained on the model for 2 classes (2 nodes at the output layer).

We have adapted the architecture of DenseNet121 for our case. Standard CNNs have the problem of Vanishing gradients when trained for deeper layers. DenseNet solves this by modifying the architecture by introducing the connectivity between the layers. Each Layer is connected to every other layer, making L*(L+1)/2 number of total connections. Connectivity concatenates the feature map instead of summing. To achieve dimensionality reduction, DenseNet is divided into different dense blocks. Using composite function operations, the output of the first layer serves as the input for the second layer. These operations are Convolution, Batch normalization and ReLU. DenseNet maintains the parameter called growth rate, which helps feature map groups in each dense block.

DenseNet121 architecture consists of 121 layers, including one 7x7 Convolution Layer, 58 3x3 Convolution layers, and 61 1x1 Convolution layers, as well as 4 AvgPool layers and 1 Fully Connected Layer at the end of the network for classification.

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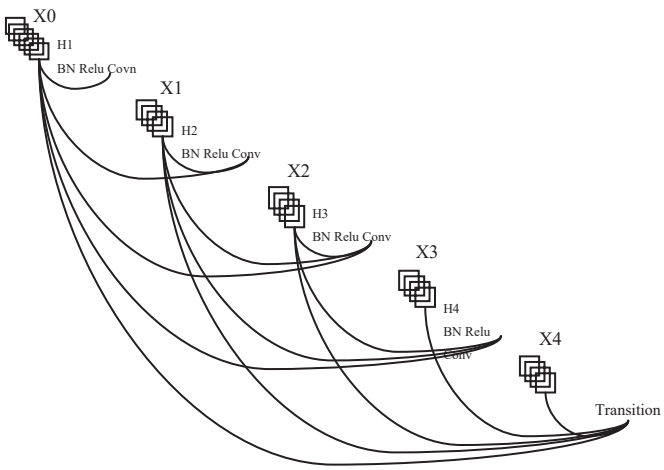

Fig. 5. DenseNet121

One addition AvgPool has been added at the bottom following the FC Dense with 1024 nodes and ReLu activation. Output layer consists two softmax activation nodes, representing each class, Real and tampered. None of the layers has been frozen as earlier weights have been trained for ImageNet classification task, which is irrelevant to our aim output.

G. Training

Dataset has been split into ratio of 80:20 for training and validation accounting total of 10091 training images and 2523 for validation. Inputs have been normalized and resized to (128,128). The model underwent training for 30 epochs, using a batch size of 32 during each epoch. The Learning Rate at the start of training is 1e-3, which is scheduled to reduce by 10% after half of the training is done.

IV. Metadata Analysis and Weather CNN

Working of the Metadata Analysis Approach has been depicted in the following block diagram.

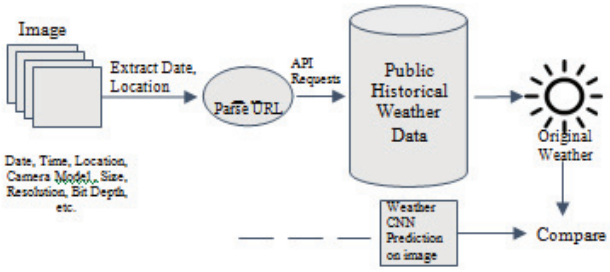

Fig. 6. Metadata Analysis Approach

A. Historical Weather API request

The system uses the metadata of an image to get the original weather conditions. Date, time, latitude and longitude are extracted from the image using EXIF. The extracted metadata is used to make an URL, which acts as an API request to open-source weather API Open-Meteo. Weather API sends back the weather condition of the location that the image is tagged with on a given date and time.

B. Model Architecture

CNN model has two convolution and two pooling layers. A convolution layer is applied with a kernel size of 1 and zero padding. This allows dimensionality reduction by reducing the number of filters and retaining important feature-related information. The fully connected layer is followed after flattening, we use 0.5 Dropout to reduce the chances of overfitting. This is followed by the last FC layer and output layer, which has 4 nodes, each representing a class of Input Images having weather conditions of Lightning, Rainy, Cloudy and Sunny.

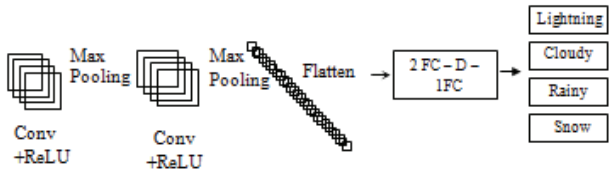

Fig. 7. Weather CNN

C. Training

Dataset has been split into ratio of 80:20 for training and validation accounting total of 1804 training images and 451 for validation. Inputs have been normalized and resized to (128,128,3). The intuition behind not grayscaling images is that we think weather conditions can correlate with pixel colour information. The model underwent training for 20 epochs, using a batch size of 32 during each epoch. The Learning Rate at the start of training is 1e-4, which is scheduled to reduce by 10% after half of the training is done.

V. Results and Discussions

A. ELA

The Proposed Error Level Analysis method has achieved maximum accuracy of 95.12%. The model has been trained for 30 epochs. Test accuracy attained using the model was 87.24%. The model took very few epochs to attain maximum accuracy because of the resulting image feature. ELA conversion helps training to be more efficient and converges the model faster. Accuracy and loss plots for training and validation can be seen below.

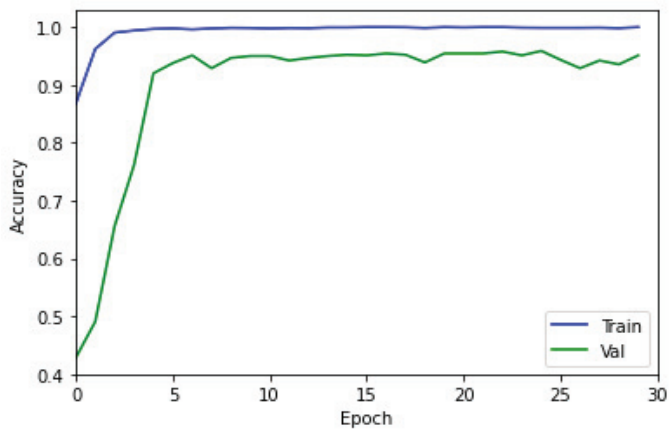

Fig. 8. (a). Val Accuracy ELA Model

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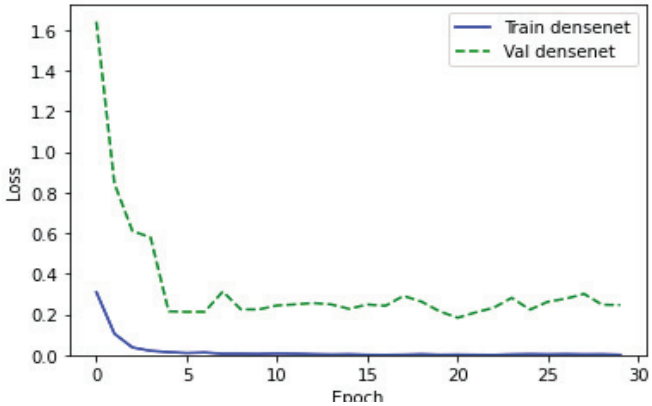

Fig. 8.(b). Val Loss ELA Model

B. *Weather CNN.*

The proposed Metadata Analysis based on weather CNN has achieved maximum accuracy of 81.60%. The model has been trained for 20 epochs. Test Accuracy attained using the model was 73.4%. Validation Accuracy and loss plots for training and validation have been depicted in the following figures

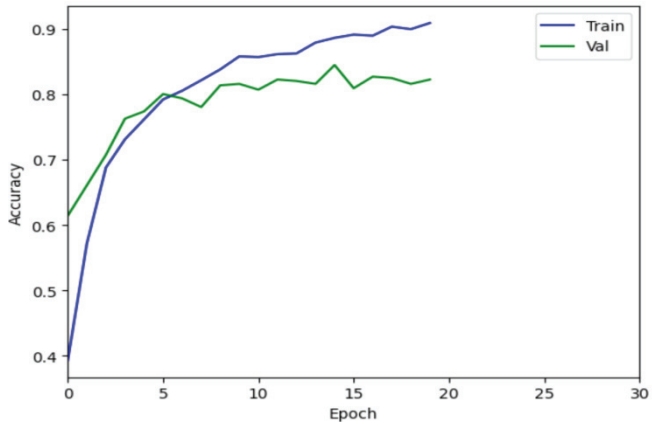

Fig. 9.(a). Val AccuracyWeather CNN

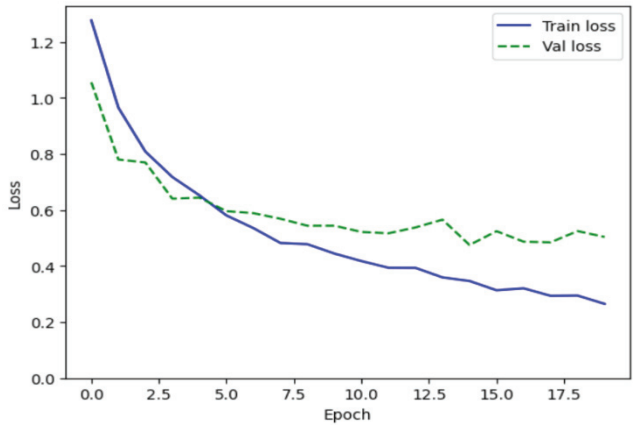

Fig. 9. (b). Val Loss Weather CNN

Overview of the Final Results of both approaches put together is depicted in the following figure:

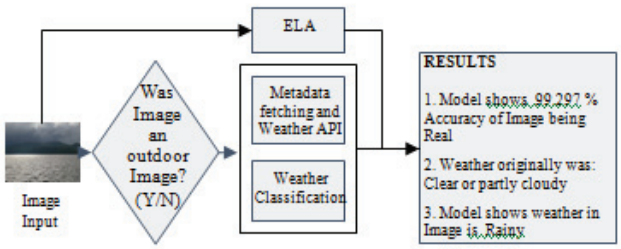

Fig. 10. Final Results

VI. CONCLUSION

In the present study, we proposed and described a method for detecting image tampering using error-level analysis and metadata analysis-based weather classification. The proposed system is capable of classifying fake and real images and has several novel features, including being more lightweight than current deep learning approaches and being more accurate than existing implementations of error-level analysis. It also combines a weather API to provide faster insights into the authenticity of an image. While the system requires images to be in lossy compression format for Error level analysis, it achieved an accuracy of 87.24% for the ELA approach and 73% for the metadata analysis approach. In future research, we will explore the integration of web traces and statistical metadata analysis to achieve more accurate and faster results, which could be useful for detecting video forgery.

I. REFERENCES

[1] V. Christlein, C. Riess, J. Jordan, C. Riess and E. Angelopoulou, "An Evaluation of Popular Copy-Move Forgery Detection Approaches," in IEEE Transactions on Information Forensics and Security, vol. 7, no. 6, pp. 1841-1854, Dec. 2012.

[2] L. Bondi, S. Lameri, D. Güera, P. Bestagini, E. J. Delp and S. Tubaro, "Tampering Detection and Localization Through Clustering of Camera-Based CNN Features," 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2017.

[3] L. Verdoliva, D. Cozzolino and G. Poggi, "A feature-based approach for image tampering detection and localization," 2014 IEEE International Workshop on Information Forensics and Security (WIFS), 2014.

[4] S. Manjunatha. and M. M. Patil, "Deep learning-based Technique for Image Tamper Detection," 2021 Third International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV), 2021.

[5] D. Cozzolino, D. Gragnaniello and L. Verdoliva, "Image forgery detection through residual-based local descriptors and block-matching," 2014 IEEE International Conference on Image Processing (ICIP), 2014.

[6] Y. Rao and J. Ni, "A deep learning approach to detection of splicing and copy-move forgeries in images," 2016 IEEE International Workshop on Information Forensics and Security (WIFS), 2016.

[7] Diallo, B., Urruty, T., Bourdon, P., Fernandez-Maloigne, C. (2019). Improving Robustness of Image Tampering Detection for Compression. In: Kompatsiaris, I., Huet, B., Mezaris, V., Gurrin, C., Cheng, WH., Vrochidis, S. (eds) MultiMedia Modeling. MMM 2019.

[8] L. Koshy and S. PraylaShyry, "Copy-Move Forgery Detection and Performance Analysis of Feature Detectors," 2020 International Conference on Communication and Signal Processing (ICCSP), 2020.

[9] C. Song, P. Zeng, Z. Wang, T. Li, L. Qiao and L. Shen, "Image Forgery Detection Based on Motion Blur Estimated Using Convolutional Neural Network," in IEEE Sensors Journal, vol. 19, no. 23, pp. 11601-11611, 1 Dec.1, 2019.

[10] P. Zhuang, H. Li, S. Tan, B. Li and J. Huang, "Image Tampering Localization Using a Dense Fully Convolutional Network," in IEEE Transactions on Information Forensics and Security, 2021.

[11] A. E. Dirik and N. Memon, "Image tamper detection based on demosaicing artifacts," 2009 16th IEEE International Conference on Image Processing (ICIP), 2009.

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[12] "L. Minh Dang, Syed Ibrahim Hassan, Suhyeon Im, Hyeonjoon Moon,Face image manipulation detection based on a convolutional neural network,Expert Systems with Applications".

[13] "Boubacar Diallo, Thierry Urruty, Pascal Bourdon, Christine Fernandez-Maloigne,Robust forgery detection for compressed images using CNN supervision,Forensic Science International: Reports"

[14] Q. Zhao, G. Cao, A. Zhou, X. Huang and L. Yang, "Image Tampering Detection via Semantic Segmentation Network," 2020 15th IEEE International Conference on Signal Processing (ICSP), 2020.

[15] B. Ambili and N. George, "A robust technique for splicing detection in tampered blurred images," 2017 International Conference on Trends in Electronics and Informatics (ICEI), 2017.

[16] R. E. Yancey, "Deep Learning for Localization of Mixed Image Tampering Techniques," 2022 29th International Conference on Systems, Signals and Image Processing (IWSSIP), 2022.

[17] D. Asatryan, N. Asatryan, N. Lanina and A. Petrosyan, "Method for detection of image tampering and partial recovery," Ninth International Conference on Computer Science and Information Technologies Revised Selected Papers, 2013.

[18] N. B. A. Warif, M. Y. I. Idris, A. W. A. Wahab and R. Salleh, "An evaluation of Error Level Analysis in image forensics," 2015 5th IEEE International Conference on System Engineering and Technology (ICSET), 2015.

[19] M. F. Jwaid and T. N. Baraskar, "Study and analysis of copy-move & splicing image forgery detection techniques," 2017 International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC), 2017.

[20] M. H. Alkawaz, M. T. Veeran and H. Razalli, "Video Forgery Detection based on Metadata Analysis and Double Compression," 2019 IEEE 7th Conference on Systems, Process and Control (ICSPC), 2019.

[21] A. V. Subramanyam and S. Emmanuel, "Video forgery detection using HOG features and compression properties," 2012 IEEE 14th International Workshop on Multimedia Signal Processing (MMSP), 2012.

[22] R. Dhanya and R. K. Selvi, "A state of the art review on copy move forgery detection techniques," 2017 IEEE International Conference on Circuits and Systems (ICCS), 2017.

[23] J. M.T, L. Koshy and G. S. Warrier, "Detection of Recoloring and Copy-Move Forgery in Digital Images," 2020 Fifth International Conference on Research in Computational Intelligence and Communication Networks (ICRCICN), 2020.

[24] T. Das, R. Hasan, M. R. Azam and J. Uddin, "A Robust Method for Detecting Copy-Move Image Forgery Using Stationary Wavelet Transform and Scale Invariant Feature Transform," 2018 International Conference on Computer, Communication, Chemical, Material and Electronic Engineering (IC4ME2), 2018.

[25] I. T. Ahmed, B. T. Hammad and N. Jamil, "Image Copy-Move Forgery Detection Algorithms Based on Spatial Feature Domain," 2021 IEEE 17th International Colloquium on Signal Processing & Its Applications (CSPA), 2021.

[26] S. D. Lin and T. Wu, "An integrated technique for splicing and copy-move forgery image detection," 2011 4th International Congress on Image and Signal Processing, 2011.

[27] H. Sasaki and Y. Kiyoki, "Automatic Adaptive Metadata Generation for Image Retrieval," 2005 Symposium on Applications and the Internet Workshops (SAINT 2005 Workshops), 2005.

[28] B. -C. Chen, P. Ghosh, V. I. Morariu and L. S. Davis, "Detection of Metadata Tampering Through Discrepancy Between Image Content and Metadata Using Multi-task Deep Learning," 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2017.

[29] Y. A. Shichkina, V. T. Tishchenko and R. R. Fatkieva, "Synthesis of the Method of Operative Image Analysis based on Metadata and Methods of Searching for Embedded Images," 2020 9th Mediterranean Conference on Embedded Computing (MECO), 2020.

[30] Z. Xiang, J. Horváth, S. Baireddy, P. Bestagini, S. Tubaro and E. J. Delp, "Forensic Analysis of Video Files Using Metadata," 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2021.

[31] E. R. Bartusiak et al., "Splicing Detection and Localization In Satellite Imagery Using Conditional GANs," 2019 IEEE Conference on Multimedia Information Processing and Retrieval (MIPR), 2019.

[32] A. Singh and J. Singh, "Image forgery detection using Deep Neural Network," 2021 8th International Conference on Signal Processing and Integrated Networks (SPIN), 2021.

[33] Sharma, P., Kumar, M. & Sharma, H. Comprehensive analyses of image forgery detection methods from traditional to deep learning approaches: an evaluation. Multimed Tools Appl (2022).

[34] "Archana V. Mire, S.B. Dhok, N.J. Mistry, P.D. Porey, Factor Histogram based Forgery Localization in Double Compressed JPEG Images,Procedia Computer Science,Volume 54,2015.

[35] Sadeghi, S., Dadkhah, S., Jalab, H.A. et al. State of the art in passive digital image forgery detection: copy-move image forgery. Pattern Anal Applic 21, 291–306 (2018).

[36] Huh, M., Liu, A., Owens, A., Efros, A.A. (2018). Fighting Fake News: Image Splice Detection via Learned Self-Consistency. In: Ferrari, V., Hebert, M., Sminchisescu, C., Weiss, Y. (eds) Computer Vision – ECCV 2018. ECCV 2018.

[37] Z. Lin, J. Sun, A. Davis and N. Snavely, "Visual Chirality," 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020.

[38] Gupta, S., Mohan, N. & Kaushal, P. Passive image forensics using universal techniques: a review. Artif Intell Rev 55, 1629–1679 (2022).

[39] Varshney, D., Vishwakarma, D.K. A unified approach of detecting misleading images via tracing its instances on web and analyzing its past context for the verification of multimedia content. Int J Multimed Info Retr 11, 445–459 (2022).

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